

PEDRAM MOHAMMADI

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PROFESSIONAL SUMMARY

- Image & video ML engineer with extensive experience shipping algorithms from research to production. Specializing in HDR/SDR tone mapping, perceptual quality optimization, video encoding, and deep learning for image & video processing. Track record of high-quality, production-ready deliverables validated by peer-reviewed publications (journals/conferences) and granted patents.

CORE SKILLS

- **ML/AI:** Vision-focused deep learning across the full lifecycle: problem framing, data curation, architecture design, large-scale training, hyperparameter tuning, rigorous validation (objective/subjective), and deployment-ready packaging.
- **Vision/Quality:** Subjective and objective image/video quality assessment, HDR/SDR tone mapping, video encoding, color processing.
- **Production:** Docker, Linux, Git, cloud ML training/inference platforms.
- **Programming & Frameworks:** Python, C/C++, MATLAB, PyTorch, OpenCV, FFmpeg.
- **Robotics Perception/SLAM:** VO/SLAM, feature matching, and detection.

PROFESSIONAL EXPERIENCE

Senior Video Quality Engineer

2021–Present

NETINT Technologies Inc., Canada

- Shipped perceptual-quality algorithms for HDR↔SDR tone mapping integrated in hardware video processing units; delivered **5–8%** gains over state-of-the-art on objective metrics.
- Led cross-block tuning and validation across VPU stages, establishing release criteria and quality gates; achieved **4%** average gain across PSNR/SSIM/VMAF test suites.
- Designed multi-granularity rate-control algorithms (frame-level, block-level, CRF, two-pass) focused on perceptual quality; achieved up to **15%** objective-metric gains on internal benches; contributions covered by granted patents.

Senior R&D Consultant

2020 - 2021

KPMG LLP, Canada

- Evaluated client R&D in image/video/algorithm design for CRA eligibility; produced technical justifications and helped unlock **\$500k–\$2M** in credits.

Software Engineer

2018–2019

TELUS Communications Inc., Canada

- Transitioned PhD research into real-time **SDR→HDR** conversion in a 1080p broadcast pipeline; validated via objective & subjective testing and deployed with measurable subscriber-visible quality improvements.

Research Scientist

2015–2020

The University of British Columbia, Canada

- Designed perception-based iTMO/TMO, hue-preserving color adjustment, image segmentation, and temporal-consistency methods; extensive subjective and objective studies demonstrated improvements over state-of-the-art baselines in visual quality; multiple IEEE publications and licensed IP.

SELECTED OPEN-SOURCE PROJECTS

Task-Aware Low-Light Enhancement for Robotics [GitHub]

- Lightweight task-aware enhancer for LoFTR matching and YOLO detection; achieved **18.45 dB PSNR** on paired validation, increased LoFTR inliers by **81%**, and improved YOLO **mAP@50** by from **10%** after adaptation on ExDark dataset.

HDR pre-processing for Visual Odometry [GitHub]

- Developed a pre-processing stage ahead of ORB-SLAM3 to improve VO robustness under extreme contrast, achieving pooled **APE** -8.8% , **RPE** -6.2% , and on TUM **APE** -10.6% with **77% win-rate**, **RPE** -12.3% against baseline.

Learned Pre-Processing for Perceptual Quality Optimization [GitHub]

- End-to-end learned preprocessing using a U-Net architecture and proxy models; **+21.2 VMAF** mean gain over **183,660** samples, enabling **30–40% bitrate reduction** at equal quality.

Video Encoder Proxy [GitHub]

- Encoder–decoder neural model trained to replicate reconstructed frames from an intra-frame encoder; achieves **30.73 dB PSNR** / **0.931 MS-SSIM** on **140k** samples and runs faster than a full encoder for research and prototyping.

VMAF Proxy [GitHub]

- Lightweight 3D CNN for VMAF prediction; achieves **PLCC 0.915**, **MAE 7.04** on **183k** samples; **2.1M** parameters for real-time content ranking/monitoring.

EDUCATION

Ph.D. in Electrical and Computer Engineering

2015 - 2020

University of British Columbia, Canada

- Thesis: High Dynamic Range (HDR) video processing
- Awards: Full graduate scholarship, International Tuition Award

M.A.Sc. in Electrical Engineering

2012 - 2014

Ferdowsi University of Mashhad, Iran

- Thesis: Subjective and Objective Image and Video Quality Assessment
- Published a widely cited survey paper on image/video quality assessment

PATENTS

- Methods and Apparatus for Video Rate Control (filed Jan 2024).
- Image processing methods, devices & media (China Patent CN116167950B, Apr 2023).
- Methods & Apparatuses for Tone Mapping and Inverse Tone Mapping (US11100888B2, Jan 2019).

PUBLICATIONS

See Google Scholar for a complete list.

WORK AUTHORIZATION

Canadian citizen.